UNIVERSITY OF ENGINEERING AND TECHNOLOGY

MARDAN

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Assignment	**Lab Assignment 11**
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Semester	**2nd**
Date	**17-04-2026**

ASSIGNMENT 11

Q no:1

Write a program that uses a single text file to write and read the provided name and age of a user. The method for implementation of writing and reading should be construction method outfile("file_name").

main.cpp > main()

```cpp
#include <iostream>
#include <fstream>
#include <cstring>
using namespace std;
int main()
{
    // for opening a file through construction method and write the data to it.
    ofstream outfile("data.txt");
    string name;
    int age;
    cout << "Enter Your name " << endl;
    getline(cin, name);
    cout << "Enter you age " << endl;
    cin >> age;
    outfile << name;
    cout << endl;
    outfile << age;
    outfile.close();
    // for opening a file fo reading data from the file
    ifstream infile("data.txt");
    string name1;
    int age1;
    getline(infile, name1);
    infile >> age1;
    cout << "Your name is " << name1 << endl;
    cout << "Your age is " << age << endl;

    infile.close();
    return 0;
```

```
PS D:\2nd semester uet mardan\OOP LABS\lzb 11\1\output> & .\main.exe
Enter Your name
ikhtisham khan
Enter you age
20
Your name is ikhtisham khan
Your age is 20
PS D:\2nd semester uet mardan\OOP LABS\lzb 11\1\output>
```

Q No 2

Create a class file_construction having character array and integer age private data attribute and two public member functions getData() and display(). The getData() method should ask user to enter name and age, write user input to a word document and close the connection. The display() function should read the word document and output the data. Create object of the class and call the necessary functions.

```
    }
    void displayData()
    {
        ifstream infile("data2.txt");
        char name1[50];
        int age1;
        infile.getline(name1, 50);
        infile >> age1;

        cout << "Name: " << name1 << endl;
        cout << "Age : " << age1 << endl;

        infile.close();
    }
};

int main()
{
    file_construction data_obj;
    data_obj.getData();
    data_obj.displayData();
    return 0;
}
```

```
PS D:\2nd semester uet mardan\OOP LABS\lzb 11\2> cd 'd:\2nd semester uet mardan\OOP LABS\lzb 11\2\output'
PS D:\2nd semester uet mardan\OOP LABS\lzb 11\2\output> & .\'main.exe'
Enter Your Name:
Ikhtisham khan
enter your age:
20
Name: Ikhtisham khan
Age : 20
PS D:\2nd semester uet mardan\OOP LABS\lzb 11\2\output>
```

File Edit View H1 B ...

```
Ikhtisham khan
20
```

Q no 3:

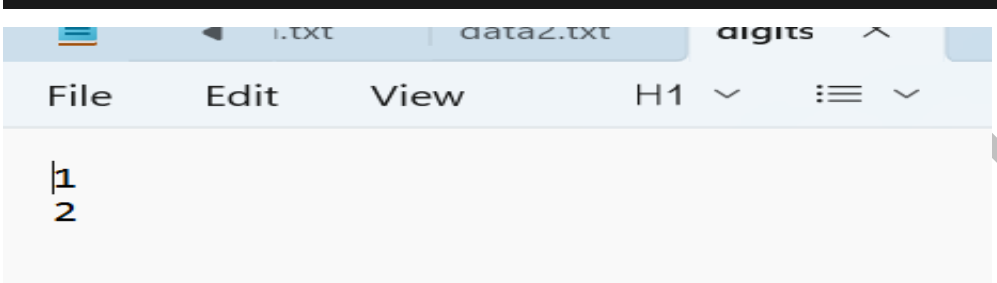
Create a text file named as digits and write the digit 1 and 2 in two lines. Read the digits text file using open() method and display both the numbers to the user

Welcome main.cpp

main.cpp > main()

```cpp
#include <iostream>
#include <cstring>
#include <fstream>

using namespace std;

int main()
{   // for writing the data into the file
    ofstream out;
    out.open("digits.txt");
    out << 1 << "\n"
        << 2 << "\n";
    out.close();
    // for reading data from the file
    ifstream in;
    in.open("digits.txt");

    int num1, num2;
    in >> num1;
    in >> num2;

    cout << "num1 " << num1 << endl;
    cout << "num2 " << num2 << endl;

    in.close();
    return 0;
}
```

```
PS D:\2nd semester uet mardan\OOP LABS\lzb 11\3> cd "d:\2nd semester uet marda
PS D:\2nd semester uet mardan\OOP LABS\lzb 11\3\output> & .\'main.exe'
num1 1
num2 2
PS D:\2nd semester uet mardan\OOP LABS\lzb 11\3\output>
```



Q no 4

Create a text file named as items and write 1 to 10 digits in alphabets in ten separate lines. Read the items text file using open() method and display the text file data to the user.

main.cpp > main()

```cpp
#include <iostream>
#include <cstring>
#include <fstream>
using namespace std;

int main()
{

    ofstream out;
    out.open("items.txt");
    out << "One\nTwo\nThree\nFour\nFive\nsix\nseven\nEight\nNine\nTen\n";
    out.close();

    ifstream in;
    in.open("items.txt");
    string item;
    while (in >> item)
    {

        cout << item << endl;
    }
    in.close();
    return 0;
}
```

```
PS D:\2nd semester uet marda
PS D:\2nd semester uet marda
One
Two
Three
Four
Five
six
seven
Eight
Nine
Ten
PS D:\2nd semester uet marda
```

2.txt | digits.txt | items

File Edit View H1

```
One
Two
Three
Four
Five
six
seven
Eight
Nine
Ten
```

ASSIGNMENT 11

Q no 5:

Create a text file named as duplicate and insert duplicate data and using open() method find out the duplicate data that is present in the file. Pretend, there is a word count that is written twice in a text file, so display the word count to the user and number of times it is repeated in the file.

main.cpp > ...

```cpp
#include <iostream>
#include <fstream>
#include <cstring>
using namespace std;

int main()
{
    ofstream out;
    out.open("duplicate.text");
    out << 1 << " " << 2 << " " << 3 << " " << 3 << " "
        << 4 << " " << 5 << " " << 3 << " " << 6 << " " << 3;
    out.close();

    ifstream in;
    int num;
    int count = 0;
    in.open("duplicate.text");
    while (in >> num)
    {
        if (num == 3)
        {
            count++;
        }
    }
    in.close();
    cout << "the number three 3 is " << count << " times in the file" << endl;
    return 0;
}
```

```
PS D:\2nd semester uet mardan\OOP LABS\lzb 11\5> cd 'd:\2nd semester uet ma
PS D:\2nd semester uet mardan\OOP LABS\lzb 11\5\output> & .\'main.exe'
the number three 3 is 4 times in the file
PS D:\2nd semester uet mardan\OOP LABS\lzb 11\5\output>
```

ts.txt items.txt dupli

File Edit View

1 2 3 3 4 5 3 6 3

Q no 6:

Read out multiple files using open() method

main.cpp

D: > 2nd semester uet mardan > OOP LABS > lzb 11 > 6 > main.cpp > main()

```cpp
#include <iostream>
#include <fstream>
using namespace std;

int main()
{
    ofstream out;
    // file 1 data
    out.open("dual.txt");
    out << 1 << " " << 2 << " " << 3 << "" << 1 << " " << 1 << " " << 1;
    out.close();
    // file 2 data
    out.open("data3.txt");
    string name = "ikhtisham khan";
    int age = 20;
    out << name << "\n"
        << age;
    out.close();
    // file three data
    out.open("digitsdata.txt");
    out << "one\n"
        << "two\nthree\nfour\nfive\n";
    out.close();
    cout << "\n*** Data from file 1 ***" << endl;
```

```
    ifstream in;
    in.open("dual.txt");
    int num;
    int count = 0;
    while (in >> num)
    {
        if (num == 1)
        {
            count++;
        }
    }
    cout << "Number of 1 in the data " << count << endl;
    in.close();
    in.clear();
    cout << "\n*** Data from file 2 ***" << endl;

    in.open("data3.txt");
    string name1;
    int age1;
    getline(in, name1);
    in >> age1;
    cout << "Name :" << name1 << endl;
    cout << "Age: " << age1 << endl;
    in.close();
    in.clear();
```

```
    cout << "\n*** Data from file 3 ***" << endl;
    in.open("digitsdata.txt");
    string item;
    while (in >> item)
    {
        cout << item << endl;
    }
    in.close();
    in.clear();

    return 0;
}
```

```
PS D:\2nd semester uet mardan\OOP LABS\lzb 11\6\output> cd 'd:\2nd seme
PS D:\2nd semester uet mardan\OOP LABS\lzb 11\6\output> & .\'main.exe'

*** Data from file 1 ***
Number of 1 in the data 3

*** Data from file 2 ***
Name :ikhtisham khan
Age: 20

*** Data from file 3 ***
one
two
three
four
five
PS D:\2nd semester uet mardan\OOP LABS\lzb 11\6\output>
```

```
ikhtisham khan
20
```

```
one
two
three
four
five
```

```
1 2 31 1 1
```

Q no 7

A file contains a list of names and telephone numbers in the following form: Name Telephone No. Write a C++ program to read the file and output the list in the tabular format. The name should be left-justified and numbers right-justified. Use a class object to store each set of data.

OOP_Lab11.cpp

OOP_Lab11.cpp > Contact > printHeader()

```cpp
#include <iostream>
#include <fstream>
#include <iomanip>
#include <string>
#include <vector>
using namespace std;

class Contact {
private:
    string name;
    string phone;

public:
    Contact(const string& n, const string& p) : name(n), phone(p) {}

    void display() const {
        cout << left  << setw(20) << name
             << right << setw(15) << phone
             << "\n";
    }

    static void printHeader() {
        cout << "\n";
        cout << left  << setw(20) << "Name"
             << right << setw(15) << "Telephone No."
             << "\n";
        cout << string(35, '-') << "\n";
```

```cpp
        cout << string(35, '-') << "\n";
    }
};

int main() {
    ifstream file("contacts.txt");

    if (!file.is_open()) {
        cerr << "Error: Could not open contacts.txt\n";
        return 1;
    }

    vector<Contact> contacts;
    string name, phone;

    while (file >> name >> phone) {
        contacts.emplace_back(name, phone);
    }

    file.close();

    Contact::printHeader();
    for (const Contact& c : contacts) {
        c.display();
    }

    cout << string(35, '-') << "\n\n";
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER powershell

```
PS D:\2nd Semester\1 OOPS\OOPS LAB\Lab Code>
cd "d:\2nd Semester\1 OOPS\OOPS LAB\Lab Code\" ; if ($?) { g++ OOP_Lab11.cpp -o OOP_Lab11 } ; if ($?) { .\OOP_Lab11 }

Name                 Telephone No.
-----------------------------------
Tahir                  03305467333
Asim                   03305567542
Ikthisham             035787899238
Ali                    03499577833
-----------------------------------

PS D:\2nd Semester\1 OOPS\OOPS LAB\Lab Code>
```

contacts - Notepad

File Edit Format View Help

```
Tahir                    03305467333
Asim                     03305567542
Ikthisham                035787899238
Ali                      03499577833
```

Q no 8:

Write an interactive, menu-driven program that will access the file created in program above and implement the following tasks:

a) To determine the telephone numbers of the specified person

b) To determine the name if a telephone number is given.

c) To update the telephone number whenever there is a change

OOP_Lab11_8.cpp > main()

```cpp
#include <iostream>
#include <fstream>
#include <iomanip>
#include <string>
#include <vector>
using namespace std;

class Contact {
public:
    string name;
    string phone;

    Contact() {}
    Contact(const string& n, const string& p) : name(n), phone(p) {}

    void display() const {
        cout << left  << setw(20) << name
             << right << setw(15) << phone
             << "\n";
    }

    static void printHeader() {
        cout << "\n";
        cout << left  << setw(20) << "Name"
             << right << setw(15) << "Telephone No."
             << "\n";
        cout << string(35, '-') << "\n";
    }
```

```
    }
};
vector<Contact> loadContacts() {
    vector<Contact> contacts;
    ifstream file("contacts.txt");

    if (!file.is_open()) {
        cerr << "Error: Could not open contacts.txt\n";
        return contacts;
    }

    string name, phone;
    while (file >> name >> phone) {
        contacts.emplace_back(name, phone);
    }

    file.close();
    return contacts;
}
void saveContacts(const vector<Contact>& contacts) {
    ofstream file("contacts.txt");

    if (!file.is_open()) {
        cerr << "Error: Could not write to contacts.txt\n";
        return;
    }

    for (const Contact& c : contacts) {
        file << c.name << " " << c.phone << "\n";
```

```

    file.close();
}
void findPhone(const vector<Contact>& contacts) {
    string target;
    cout << "\nEnter name to search: ";
    cin >> target;

    bool found = false;
    for (const Contact& c : contacts) {
        if (c.name == target) {
            cout << "\nPhone number of " << c.name << ": " << c.phone << "\n";
            found = true;
            break;
        }
    }

    if (!found)
        cout << "\nNo contact found with name: " << target << "\n";
}
void findName(const vector<Contact>& contacts) {
    string target;
    cout << "\nEnter telephone number to search: ";
    cin >> target;

    bool found = false;
    for (const Contact& c : contacts) {
        if (c.phone == target) {
```

```
        if (c.phone == target) {
            cout << "\nName for " << c.phone << ": " << c.name << "\n";
            found = true;
            break;
        }
    }

    if (!found)
        cout << "\nNo contact found with number: " << target << "\n";
}
void updatePhone(vector<Contact>& contacts) {
    string target;
    cout << "\nEnter name whose number you want to update: ";
    cin >> target;

    bool found = false;
    for (Contact& c : contacts) {
        if (c.name == target) {
            cout << "Current number: " << c.phone << "\n";
            cout << "Enter new number: ";
            cin >> c.phone;
            saveContacts(contacts);
            cout << "Updated successfully.\n";
            found = true;
            break;
        }
    }
```

```
    if (!found)
        cout << "\nNo contact found with name: " << target << "\n";
}
void displayAll(const vector<Contact>& contacts) {
    Contact::printHeader();
    for (const Contact& c : contacts) {
        c.display();
    }
    cout << string(35, '-') << "\n";
}
void printMenu() {
    cout << "\n========== CONTACT BOOK ==========\n";
    cout << "1. Show all contacts\n";
    cout << "2. Find phone number by name\n";
    cout << "3. Find name by phone number\n";
    cout << "4. Update a phone number\n";
    cout << "0. Exit\n";
    cout << "===================================\n";
    cout << "Enter choice: ";
}
int main() {
    vector<Contact> contacts = loadContacts();

    if (contacts.empty()) {
        cout << "No contacts loaded. Exiting.\n";
        return 1;
    }

    int choice;
```

```
    int choice;

    do {
        printMenu();
        cin >> choice;

        switch (choice) {
            case 1: displayAll(contacts);      break;
            case 2: findPhone(contacts);       break;
            case 3: findName(contacts);        break;
            case 4: updatePhone(contacts);     break;
            case 0: cout << "\nGoodbye.\n";    break;
            default: cout << "\nInvalid option. Try again.\n";
        }

    } while (choice != 0);

    return 0;
}
```

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   at Microsoft.PowerShell.PSConsoleRe
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   at Microsoft.PowerShell.PSConsoleRe
-------------------------------------
_Lab11_8.cpp -o OOP_Lab11_8 } ; if ($?) { .\OOP_Lab1
1_8 }

========== CONTACT BOOK ==========
1. Show all contacts
2. Find phone number by name
3. Find name by phone number
4. Update a phone number
0. Exit
===================================

Name                  Telephone No.
-----------------------------------
Tahir                   03305467333
Asim                    03305567542
Ikthisham              035787899238
Ali                     03499577833
-----------------------------------

========== CONTACT BOOK ==========
1. Show all contacts
2. Find phone number by name
3. Find name by phone number
4. Update a phone number
0. Exit
===================================
Enter choice: 2

Enter name to search: Ali
```

```
========== CONTACT BOOK ==========
1. Show all contacts
2. Find phone number by name
3. Find name by phone number
4. Update a phone number
0. Exit
===================================
Enter choice: 2

Enter name to search: Ali

Phone number of Ali: 03499577833

========== CONTACT BOOK ==========
1. Show all contacts
2. Find phone number by name
3. Find name by phone number
4. Update a phone number
0. Exit
===================================
Enter choice: 0

Goodbye.
```